Technology Stack

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Our Technology Stack:
The proposed Flood Prediction System uses Python, Flask, HTML, CSS, and JavaScript for application development, along with Scikit-learn, XGBoost, Pandas, and NumPy for machine learning and data processing. The solution is designed to be scalable, maintainable, and suitable for deployment on IBM Cloud for real-time flood prediction.

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, JavaScript, Bootstrap	Used to build a responsive and user-friendly web interface for entering weather parameters and displaying flood prediction results.
2	**Backend / Server-Side**	Python, Flask	Flask provides a lightweight web framework for integrating the machine learning model and handling prediction requests efficiently.
3	**Database / Data Storage**	CSV Dataset (Kaggle), Joblib (.pkl), SQLite (Optional)	Stores the historical flood dataset, trained machine learning model, and prediction records for efficient retrieval and future analysis.
4	**Cloud / Hosting / Deployment**	IBM Cloud, Flask Development Server	IBM Cloud enables scalable deployment of the application, allowing users to access flood prediction services remotely.
5	**Version Control & CI/CD**	Git, GitHub	Git provides version control, while GitHub enables collaborative development, source code management, and project tracking.

6	**Third-Party APIs / Other Tools**	Scikit-learn, XG-Boost, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	Used for data preprocessing, visualization, machine learning model training, evaluation, and development of the flood prediction system.